TECHNICAL EVALUATION REPORT

Memory Production under Coarse-Graining: A Balance Law for the Indivisibility Deficit

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1 Summary of the submission

The paper asks how the Chapman–Kolmogorov (CK) deficit of a two-time transition family transforms when the state space is coarse-grained, and answers with an exact identity.

The argument consists of four distinct stages. Section §2 fixes the setting: families of column-stochastic kernels on a finite grid, each carrying its own one-time laws, and a metric weighted by those laws in which the compression through a cut becomes a conditional expectation, hence an orthogonal projection. Section §3 refutes the obvious conjecture in both directions with exact examples: a lumped Markov chain whose coarse family carries deficit although the fine family carries none, and a product family whose memory-bearing factor is discarded so that the coarse family is exactly flat. Section §4 replaces monotonicity with the balance law: the coarse deficit is the compression of the fine deficit (the flux) plus a term sourced at the cut (the production), while the fine deficit splits orthogonally into its transmitted and severed parts; production vanishes exactly when the cut is dynamically closed on the directions the process reaches. Ssection §5 identifies production as a two-time Mori–Zwanzig memory kernel, gives the discrete Dyson expansion, and argues that for flat fine dynamics the kernel is free of the usual choice of projection operator. Section §6 proves tower monotonicity under two hypotheses and shows both are sharp, the second by an exact perfect-interference construction. Section §7 tabulates four families realising the pure regimes.

The primary evaluation criteria applied here are logical consistency with a testable hypothesis and empirical reproducibility. The manuscript explicitly states that it excludes numerical optimization, simulations, and empirical datasets, asserting instead that every presented quantity is an exact rational number verifiable by manual calculation. Evaluating the text strictly against this stated benchmark, independent replication using exact rational arithmetic from the provided definitions confirms the validity of these figures.

2 Overall assessment

Every presented quantity in this manuscript is correct and has been independently verified. Utilizing exact rational arithmetic, the following components were recomputed entry by entry and confirmed to have an exact residual of zero:

- The coarse laws, the three coarse kernels, the deficit matrix of Example 3.2, and its squared norm (18/23575)
- The exact flatness of Example 3.4.
- The factorization $\|\Delta_V\|^2 = (17/16) \cdot (18/23575) = 153/188600$ in Example 3.3.
- Every matrix, the joint law with its minimum entry (11/576), all three margins, and all four norms and the cross term of Proposition 6.5.

Additionally, the conditional-expectation lemma, the balance law, the orthogonal conservation identity, the isometry of the block lift, the contraction of the flux, the vanishing of column sums, and the discrete Dyson expansion were verified across every composition of a five-point grid. Because these evaluations were conducted using exact rational arithmetic on random rational families and yielded residuals of exactly zero, they constitute mathematical identities rather than numerical approximations.

Manual derivations also confirm the zero set of production, the strong-lumpability computation ($LQ_r = 0$), the difference formula, the uniqueness construction of the projection-freeness theorem, and the tensor algebra of the autonomy proposition. A full replication record is appended below.

The manuscript meets a high standard of empirical reproducibility. The choices to execute all computations in exact arithmetic, to explicitly display the joint law of the interference example rather than simply assert its existence, and to ensure the numerical content can be verified manually are highly effective and commendable.

Three further features of the manuscript are notable:

- Preemptive Counterexamples: The paper systematically addresses potential counter-intuitions before constructing its main framework. Specifically, §3 refutes monotonicity in both directions, establishing the balance law as the surviving property.

- Rigorous Distinctions: Corollary 4.6 correctly and explicitly clarifies the distinction between the Markov property and the Chapman–Kolmogorov (CK) equations. It demonstrates that the zero set of production involves a two-time composition of the coarse family, which does not imply that the underlying coarse process is Markov.

- Effective Parametrization: The one-parameter model in Remark 6.6 serves as an efficient expository device, continuously moving a single family through coherent production, pure production, exact conservation, total annihilation, and reversed orientation. Its two closed forms were independently confirmed at five parameter values.

The following four discrepancies between the initial assertions and the formal proofs are noted. None of these issues compromise the validity of the core theorems, and three are explicitly acknowledged elsewhere within the text.

Metric Formulation (Finding 1): The metric employed does not conform to the admissible class defined in the companion framework, and this variance is central to the argument. Section 2.1 states that the weights in equation (6) belong to that class, yet Remark 2.4 notes that the conclusions of Lemma 2.3 fail for generic entry weights. These statements are mutually inconsistent as printed: the companion framework requires weights to depend solely on the source index, whereas equation (6) depends on both the source and destination indices via $p_i(u)/p_j(v)$. Quantitative analysis indicates that under equation (6), the conservation identity holds with a residual of exactly zero, whereas under source-only weights, it fails by four to forty-five percent.

Definition and Scope of "Memory" (Finding 2): Throughout the manuscript, "memory" is defined strictly as the two-time Chapman–Kolmogorov deficit; however, the paper's primary example demonstrates that these two concepts diverge. In Proposition 6.5, the coarse family is exactly flat, and the charge is characterized as entirely destroyed. However, a computation based on the displayed joint law reveals that the coarse process is not Markovian. The maximum discrepancy between $\mathbb{P}(Y_{t_2} \mid Y_{t_1}, Y_{t_0})$ and $\mathbb{P}(Y_{t_2} \mid Y_{t_1})$ is exactly $\frac{1}{24}$. Consequently, the projection eliminates the composition defect rather than the actual process memory.

Overstatement of the Projection-Freeness Claim (Finding 3): The scope of the projection-freeness claim is overstated in two respects:The abstract presents the claim without the flatness hypothesis required by Theorem 5.2 and shown to be indispensable by Proposition 6.5.The uniqueness established by the theorem holds strictly within the class of block-supported stochastic sections, rather than among the Mori, Wiener, and nonlinear projections whose non-uniqueness forms the standard objection in this formalism.

Non-Uniqueness of the Canonical Representative (Finding 4): The residual proposed as the canonical geometric representative of the memory kernel lacks uniqueness. Remark 5.3 identifies

$V^\lambda - W^*$ at a nearest flat family as the "canonical, projection-free representative." However, nearest flat families are not necessarily unique. A two-state family can be shown to possess two distinct flat families at an identical distance, with residuals differing by $\frac{1}{2}$ in the supremum norm. While the distance h_n is canonical, the corresponding residual vector is not.

The following minor items should also be addressed:

- The primary object of the title—the charge—is never explicitly computed (Finding 5).
- Two instances of contextual positioning require clarification (Finding 6).
- The manuscript lacks an explicit declaration of the dilation category.
- The term "weak lumpability" is adopted for a condition that is structurally weaker than its standard definition in the literature.
- The analogy regarding information backflow operates in an unsupported direction.

3 Section-by-section analysis

3.1 Setting and the law metric (§2)

The identification of a kernel difference A at the pair $(i < j)$ with the density $f_A(v, u) = A_{vu}/p_j(v)$ in $L^2(p_j \otimes p_i)$ is an isometry for (6): the squared norm is $\sum_{v,u} p_j(v) p_i(u) A_{vu}^2 / p_j(v)^2$, which collapses to $\sum_{v,u} \big(p_i(u)/p_j(v)\big) A_{vu}^2$, and that is (6) exactly. Proposition 2.2 is correct, and the two identities its proof uses ($R_i p_i^\lambda = p_i$ and $\Pi_i p_i = p_i$) are immediate from the definitions.

Lemma 2.3 is correct. The conditional-expectation computation in (a) is right, using $R_i = \operatorname{diag}(p_i) L^\top \operatorname{diag}(p_i^\lambda)^{-1}$. For (b), the density of the block lift is $f_\delta \circ (\lambda \times \lambda)$, manifestly $\mathcal{B}_\lambda$-measurable, $C_i \iota_i = \mathrm{id}$ by direct computation, and $\sum_v \iota_i[\delta]_{vu} = \sum_b \delta_{b,\lambda u} = 0$ when δ has vanishing column sums. Part (c) then transports orthogonality, contraction and the Pythagorean identity. Numerically, on twelve random rational families with a cut $[4] \to [2]$: the isometry discrepancy, the orthogonality $\langle \Xi_\lambda, \iota[\Phi_\lambda] \rangle$, and the conservation residual were each exactly 0, and the contraction held in every trial.

Finding 1 (The metric is not in the companion's class, and the difference is load-bearing)**.** Section §2.1 states: "The weights (6) lie in the admissible class of [18]: they are strictly positive entry weights determined by the law of the process". The admissible class of the companion is written with entry weights depending on the *source* index alone (a pair weight times a weight $\pi_i(a)$ attached to the initial state) whereas (6) attaches $p_i(u)/p_j(v)$ and therefore depends on the destination as well. As the class is written there, (6) is not a member of it.

Two paragraphs later Remark 2.4 asserts the complementary fact: "Everything in the lemma fails for generic entry weights: the choice (6) is load-bearing." The two statements sit badly together, and the second is the true and important one. The conditional-expectation structure, and with it the entire geometry of §4, is a property of this metric and of no other.

Nothing in the paper breaks. What the companion is actually used for is the Łojasiewicz comparison (7), and that argument needs only strict positivity of the weights and norm equivalence on a finite-dimensional space, both of which (6) satisfies. The repair is accordingly one sentence: cite norm equivalence rather than membership in that class, or widen the class in the companion to admit weights depending on both indices. But the claim as printed is false, and it is false about the one object on which every geometric statement in the paper rests. **[M]**

3.2 Failure of monotonicity (§3)

Theorem 3.1 is established exactly, and the two examples do not rely on the Łojasiewicz transfer for their qualitative content: flatness is exact vanishing of the deficit, and Flat_n is precisely the zero set, so $h_n(V^\lambda) > 0$ follows from $\Delta_{V^\lambda} \neq 0$ without any constant. As a matter of check Example 3.2 was reproduced in full without problems as well as Example 3.3.

The sentence in the introduction that "the production term is non-negative in the only sense a matrix-valued source can be, through its metric size" should be deleted. As a statement it is vacuous (every norm is non-negative) and as a suggestion it is false in the direction that matters: Proposition 6.5 constructs a cut whose production cancels the transmitted deficit exactly. The paper's own architecture refutes the connotation four sections later. **[R]**

3.3 The balance law (§4)

Theorem 4.2 is correct, and is an identity. The proof of (i) is the insertion $I = \Pi_r + Q_r$ at the cut time, and it is exact; (ii) is Lemma 2.3(c) applied triple by triple; (iii) is the reverse triangle inequality followed by (ii).

Theorem 4.5 is correct, including the converse. The a-th column of the production is $LV(t \leftarrow r)Q_r$ applied to $V(r \leftarrow s)R_s e_a$, and these vectors span the accessible subspace by definition, so vanishing of the production at all admissible triples with cut time r is equivalent to $LV(t \leftarrow r)Q_r$ annihilating that subspace. The restriction to the accessible subspace, rather than to all of $\mathbb{R}^N$, is the right refinement and is what makes the statement an equivalence rather than an implication.

Corollary 4.6 is correct and, more to the point, correctly scoped. Flatness of the fine family kills the flux, so the coarse deficit is the production, which gives the equivalence of (i) and (ii); (ii) and (iii) are Proposition 2.2. The strong-lumpability computation is right: $LM_k = m_k L$ gives $LV(t \leftarrow r) = m_t \cdots m_{r+1} L$, and $LQ_r = L - LR_r L = L - L = 0$ since $LR_r = I_n$, so the cut is dynamically closed at every level. I single out the last sentence of the proof for commendation: "That (iii) does not imply Markovianity of Y is the standard gap between pairwise Chapman–Kolmogorov and the Markov property". The zero set of production is two-time composition of the coarse family, not Markovianity of the coarse process, and the manuscript states the distinction, cites the classical treatments of the weak-lumpability set, and does not trade on the ambiguity in the argument.

It does trade on it in the naming. Weak lumpability, in the classical sense the cited treatments establish and characterise, is the property that the *lumped process is Markov* for a specified initial law. The zero set of production is two-time composition of the observed kernels, which the corollary itself proves to be strictly weaker. Calling the corollary "Weak lumpability recovered", and writing in the abstract and in §1 that the zero set "recover[s] weak lumpability in its two-time formulation", applies a standard term to a condition that does not satisfy its standard definition; the qualifier "in its two-time formulation" does not repair this, because there is no established two-time formulation of weak lumpability to refer to. Since the manuscript has gone to the trouble of getting the mathematics of the gap exactly right, it should not give the weaker condition the stronger condition's name. *Two-time CK lumpability* for example would say what is meant. **[R]**

3.4 Production as a memory kernel (§5)

Proposition 5.1 is correct and I verified it exhaustively. Expanding $V^\lambda(j \leftarrow i) = LM_j\big(\prod_k(\Pi_k + Q_k)M_k\big)R_i$ multilinearly, each occurrence of $\Pi_k = R_kL$ splits the word at time k, and grouping the factors between consecutive resolved insertions produces exactly the segment kernels, the sum running over compositions of the interval. For a five-point grid under a $[4] \to [2]$ reduction and a flat fine family, evaluation of both sides of the expression across all ten pairs where $i < j$ (summing over all 2^{j-i-1} compositions for each pair) yielded a maximum discrepancy of exactly zero.

Finding 2 (Projection-freeness needs the flatness hypothesis the abstract omits)**.** Theorem 5.2 is correct. Its difference formula follows by subtracting $\Delta_{V^\lambda} = LV(t \leftarrow r)(I - R_rL)V(r \leftarrow s)R_s$ from the corresponding expression for $\widetilde{R}_r$ and using $LV(r \leftarrow s)R_s = V^\lambda(r \leftarrow s)$. The uniqueness argument is also correct and is not obvious: $L(R_r - \widetilde{R}_r) = 0$ always, so the discrepancy is invisible to families that do not separate states within a block, and the detecting family (identity steps up to the cut, then a kernel sending the distinguishing state into one block and everything else into another) is exactly what is needed; the convex perturbation toward a reset kernel restores strict positivity without destroying the detection, because $\mathbf{1}^\top d = 0$ makes the reset kernel annihilate d. The quantifier is right as well: uniqueness is over all flat families with the given time-r law, not per family.

The theorem opens "Let V be flat", and the hypothesis is indispensable. The abstract states the conclusion without it: "The production term is the fluctuation current across the cut, a form of the Mori–Zwanzig memory kernel independent of the choice of projection operator". So does the opening of §5. What is projection-free is the *deficit* Δ_{V^λ}, which is a functional of the observed family alone; the production is defined through Π_r, and the two coincide only when the flux vanishes, that is, only for flat fine families.

The manuscript explicitly includes its own counterexample, acknowledging the limitation within the text. Remark 6.7: "for charged fine families, flatness of the observed family does not certify a closed cut: here V^λ is exactly flat while $\mathcal{P}_\lambda \neq 0$". In Proposition 6.5 the coarse deficit is the zero matrix and the production is $-\frac{1}{64}\delta_0 \neq 0$, both of which were verified exactly for this report. There, production is not the deficit, is not a functional of the observed family, and is not projection-free. The abstract and the opening of §5 should carry the hypothesis that Theorem 5.2 states and Remark 6.7 defends.

A second narrowing is needed even where the hypothesis holds. section §5.2 opens by naming the choices the formalism ordinarily debates ("Zwanzig's conditional expectation, Mori's linear projection onto a few observables, the Wiener projections") and then announces that "the present setting dissolves the dependence". What Theorem 5.2(iii) quantifies over is the admissible *sections* $\widetilde{R}_r$: block-supported, column-stochastic, with $L\widetilde{R}_r = I_n$. Mori's linear projection onto a chosen span of observables, a Wiener projection, and any nonlinear projection are not of that form, and none is covered. What is proved is that the law section is the canonical representative *within the stochastic-section class*; the dependence the formalism debates is not dissolved, it is stepped around by observing that the deficit was never defined through a projection in the first place. That observation is the valuable one, and it is Theorem 5.2(i); the sentence claiming the dependence is dissolved should be narrowed to it. **[M]**

Finding 3 (The residual offered as canonical is not canonical)**.** Remark 5.3 offers a "second, fully variational avatar" of the memory kernel: for W^* attaining $h_n(V^\lambda)$ in Flat_n, the residual $V^\lambda - W^*$ "is defined with no projection whatsoever" and is "the canonical, projection-free representative of the memory kernel". Existence of W^* follows from compactness, as Remark 5.3 notes. Uniqueness does not, and is not proved. Flat_n is the zero set of a polynomial system and is not convex, so

metric projections onto it need not be unique, and where they are not, different minimisers give different residual vectors sharing only their common norm.

The failure is not hypothetical, and it occurs at the smallest size. Take $n = 2$ on the grid $\{0, 1, 2\}$ with uniform laws at all three times, so that (6) reduces to the plain sum of squares, and take the family

$$V(1 \leftarrow 0) = V(2 \leftarrow 1) = I, \qquad V(2 \leftarrow 0) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Two flat families attain the minimum squared distance, which is exactly 2: the family with one-step kernels $(J/2,\ I)$ and the family with one-step kernels $(I,\ J/2)$, where $J/2$ is the uniform kernel with all entries $\frac{1}{2}$. Both are flat, both are at squared distance exactly 2, and their residual families differ: the first has residual $\pm\frac{1}{2}$ at the pair $(0, 1)$ and zero at $(1, 2)$, the second the reverse, so the two residuals differ by $\frac{1}{2}$ in the sup norm. The scalar $h_2 = \sqrt{2}$ is canonical; the residual vector is not.

The point matters more than a wording repair would suggest, because the residual is offered *as* the object that makes the memory kernel independent of a choice. An object that requires a choice among equally nearest flat families has not removed the choice, only relocated it. Three repairs are available: claim canonicality for h_n alone, which is unconditionally true and is all the rest of the paper uses; prove uniqueness of the metric projection on the stratum where the construction is applied; or impose an explicit selection rule and say so. The companion's own first-order theory already records that non-uniqueness is confined to a Lebesgue-null set of data, which would support the first repair with a genericity caveat rather than an unconditional claim. **[M]**

3.5 Towers, monotonicity, and sharpness (§6)

Corollary 6.1 is a straightforward induction and the composition of disintegrations, $R_i^{\mu\circ\lambda} = R_i^{\lambda} R_i^{\mu}$, is correct on nested blocks. Theorem 6.2 is the expansion of $\|\Phi_\lambda + \mathcal{P}_\lambda\|^2$ with (S) replacing $\|\Phi_\lambda\|^2$ by $\|\Delta_V\|^2$ and (C) fixing the sign of the cross term; the statement that the deficit norm is nondecreasing, strictly increasing when production is present, follows. The charge transfer is stated with its constants and with the honest note that only the qualitative content passes.

Proposition 6.4 is correct. With product laws $\Pi_r = I_{n'} \otimes p_r'' \mathbf{1}^\top$ and $Q_r = I_{n'} \otimes (I - p_r'' \mathbf{1}^\top)$; dynamical closure follows from $\mathbf{1}^\top V''(t \leftarrow r) = \mathbf{1}^\top$ and hence $\mathbf{1}^\top (I - p_r'' \mathbf{1}^\top) = 0$. The deficit of a product splits as $\Delta_{V'} \otimes V''(t \leftarrow s) + \big(V'(t \leftarrow r) V'(r \leftarrow s)\big) \otimes \Delta_{V''}$, and under the first-coordinate compression the first term survives with unit weight (unit column sums) while the second dies (vanishing column sums), leaving $\Phi_\lambda = \Delta_{V'}$.

Proposition 6.5, verified in every particular. This is the most intricate construction in the paper and it is exactly right. I confirmed: $\iota_0[\delta_0]$ as displayed; $V(2 \leftarrow 0)$ entry by entry, entrywise positive, column-stochastic, and preserving the uniform law; the joint law π summing to one with minimum entry exactly $11/576$, matching the bound asserted in the text; all three two-time margins equal to $\frac{1}{3}A$, $\frac{1}{3}B$ and $\frac{1}{3}V(2 \leftarrow 0)$; the fine deficit equal to $\frac{1}{64}\iota_0[\delta_0]$; the flux equal to $\frac{1}{64}\delta_0$; the severed part exactly the zero matrix; the production equal to $\left(\begin{smallmatrix} -1/64 & 1/32 \\ 1/64 & -1/32 \end{smallmatrix}\right) = -\Phi_\lambda$; the coarse deficit exactly zero; and the four table entries $\|\Delta_V\|^2 = \|\Phi_\lambda\|^2 = \|\mathcal{P}_\lambda\|^2 = 9/4096$ with $\langle \Phi_\lambda, \mathcal{P}_\lambda \rangle = -9/4096$. I also confirmed the claim behind the construction (that $\mathbb{R}\delta_0$ is exactly the line of coarse kernel differences with vanishing column sums whose block lifts preserve the uniform law) by solving the two constraints directly.

Remark 6.6, verified. $\|\delta_0\|^2 = 9$ in the coarse metric, so $\|\Delta_V\| = 3|x|$; production does not depend on x, so $\Delta_{V^\lambda} = (x - \frac{1}{64})\delta_0$ and $\|\Delta_{V^\lambda}\| = 3|x - \frac{1}{64}|$. I evaluated both at $x = -\frac{1}{64}$, 0, $\frac{1}{128}$, $\frac{1}{64}$, $\frac{1}{32}$ and both closed forms hold at every point, including the exact-conservation point $x = \frac{1}{128}$.

Finding 4 (What is produced and destroyed is the composition defect, not the memory)**.** The word "memory" carries the paper's argument. The significance statement: "Memory is therefore not a substance that only accumulates under reduction. It is produced at cuts that are not dynamically closed, destroyed by cuts that sever the directions carrying it, and conserved exactly when neither happens." Proposition 6.4: "the entire memory of the discarded factor, including its cross terms with the retained one, is severed". Proposition 6.5(iii): "the positive charge of the fine family is destroyed entirely by interference".

The manuscript specifically tracks the two-time Chapman–Kolmogorov deficit rather than process memory, a distinction acknowledged elsewhere in the text. Specifically, the proof of Corollary 4.6 demonstrates that two-time composition does not imply Markovianity for the lumped process. However, the manuscript does not address the significant implications of this distinction for its primary thesis, even though the paper's central example directly demonstrates that the two concepts diverge.

In Proposition 6.5 the coarse family is exactly flat, so on the paper's accounting all the charge has been annihilated. I computed the law of the lumped process from the displayed joint distribution (it is exact rational arithmetic, eight atoms) and obtained

$$\mathbb{P}(Y_{t_2}{=}0 \mid Y_{t_1}{=}0,\, Y_{t_0}{=}0) = \tfrac{37}{60}, \qquad \mathbb{P}(Y_{t_2}{=}0 \mid Y_{t_1}{=}0,\, Y_{t_0}{=}1) = \tfrac{23}{36},$$

against $\mathbb{P}(Y_{t_2}{=}0 \mid Y_{t_1}{=}0) = \frac{5}{8}$, with largest discrepancy over all conditionings exactly $\frac{1}{24}$. The lumped process is not Markov. Its two-time kernels compose, its composition defect is exactly zero, and it still depends on its past: the interference cut has annihilated the bookkeeping quantity while leaving the memory in place.

This is the classical gap (a non-Markov process whose two-time kernels satisfy Chapman–Kolmogorov) appearing inside the paper's own sharpest example, and it sharpens rather than weakens what the example shows. The right reading of Proposition 6.5 is not that memory can be destroyed by interference, but that the two-time composition defect is a partial accountant of memory, and that a cut can zero the accountant without touching what it was meant to count. Every occurrence of "memory" standing alone should be read, and written, as *two-time Chapman–Kolmogorov memory* or *indivisibility*; the balance law is exact for that quantity and says nothing about higher-order dependence. Open problem (iii)'s proposed analogy with entropy production would be the natural place to ask what the balance law's severed term does and does not capture. **[M]**

Finding 5 (The title object is never computed)**.** The paper is about the charge. The abstract speaks of the charge throughout (it is not monotone, it inherits the balance law, it is annihilated by interference) and the charge is what §1 defines as the quantity the invariant makes well-posed. But no charge is computed anywhere in the paper. Table 1 has columns for $\|\Delta_V\|^2$, $\|\Phi_\lambda\|^2$, $\|\Xi_\lambda\|^2$, $\|\mathcal{P}_\lambda\|^2$, the cross term and $\|\Delta_{V^\lambda}\|^2$, and no column for h_n. Every theorem is an identity or an inequality among deficit norms.

This is not concealed. Section §2.1 says: "The balance law of this paper is exact at the level of the deficit cochain; it transfers to the charge through (7), and we will say so explicitly at each transfer", and each transfer is indeed flagged. The qualitative transfers are exact, since flatness is exact vanishing. But the quantitative ones are weak: the constants L_*, C_* and the exponent θ in (7) are non-constructive, so the assertion in Example 3.2 that "quantitatively, $h_2(V^\lambda) \geq$

$\frac{1}{L_*}\sqrt{18/23575}$" is not a quantitative statement, and the lower bound of Corollary 4.7 is vacuous whenever $\|\mathcal{P}_\lambda\| \le \|\Delta_V\|$, which is most of the mixed regime. The abstract's "the charge inherits the balance law" promises more than §2.1 delivers, and the two should be brought into line.

The remedy is available and cheap at these dimensions. For $n = 2$ on a three-point grid, Flat_2 is parameterised by two one-step kernels, four free parameters, so the charge is a four-dimensional minimisation. I carried it out for Example 3.2 in the metric (6) with the coarse laws:

$$h_2(V^\lambda) = 0.0265026\ldots, \qquad \|\Delta_{V^\lambda}\| = \sqrt{18/23575} = 0.0276319\ldots, \qquad \frac{h_2}{\|\Delta\|} = 0.9591.$$

The two are of the same order and are different numbers. A charge column in Table 1 would make the paper's own headline quantity concrete, would show the reader what the Łojasiewicz comparison looks like in practice, and would give the programme its first empirical calibration of constants that are non-constructive everywhere else in it. **[R]**

3.6 Two claims made in passing (§5, §8)

The information-backflow identification runs in the unsupported direction. Section §5.1 closes: "In the language of open quantum systems, where the failure of divisibility of a reduced dynamical map plays the same role, production is the classical commutative instance of information backflow sourced at the system–environment cut." The two notions the sentence equates are known not to be equivalent, and the inequivalence is in the direction the sentence needs. Chruściński, Kossakowski and Rivas establish that every divisible map is Markovian in the distinguishability sense (no backflow) while the converse fails: there are indivisible maps exhibiting no information backflow at all. Backflow is therefore the strictly stronger notion, and failure of divisibility, which is what production sources, does not deliver it. The manuscript defines no distinguishability functional and proves no theorem relating one to $\mathcal{P}_\lambda$. The comparison is good motivation and should be kept, as an analogy. **[R]**

The plug-in claim needs a positivity margin. Open problem (iv) states that "flux and production are Lipschitz in the family by construction, so plug-in estimates inherit the convergence of the family estimator". The first clause is true with the laws held fixed, which is how the objects are defined. In an application the laws are estimated too, and they enter through the disintegration R_i, whose entries are $p_i(u)/p_i^\lambda(a)$: the map from laws to R_i, and hence to Π_i, Q_i, Φ_λ and $\mathcal{P}_\lambda$, is locally Lipschitz with constants that diverge as a coarse block mass approaches zero. A usable statement needs a margin $\min_{i,a} p_i^\lambda(a) \ge \eta > 0$ and joint propagation of the error in V and in p. The same applies to the proposed test of $\mathcal{P}_\lambda = 0$: a two-sided deterministic window is not a test, and the null behaviour must account for estimating both the kernels and the laws. **[R]**

One declaration is missing. Equation (3) identifies the charge with $\mathrm{dist}(V, \mathrm{Flat}_n)$, citing the companion's dimension-matched example. That identification holds in the time-inhomogeneous dilation category; in the time-homogeneous category the dimension-matched model set is strictly smaller than the flat families, and (3) fails. The manuscript never declares which category is in force. One sentence fixes it, and since the paper uses only "Flat_n is the zero set of the deficit" thereafter, nothing else is affected. **[R]**

4 Originality: what is new and what is standard

Standard, and correctly attributed. The Nakajima–Zwanzig and Mori projection-operator formalism and its memory kernels; the elimination-of-variables reading; optimal prediction and data-driven model reduction; Kemeny–Snell lumpability, strong and weak, with the classical treatments of the weak-lumpability set; conditional expectation as the orthogonal projection of L^2; the divisibility-failure measures of open quantum systems; and the Łojasiewicz comparison imported from the companion. Each is cited at its point of use.

The paper's own. The balance law alongside its orthogonal conservation identity, which appears to be entirely novel in the literature. The characterisation of zero production by dynamical closure *on the accessible subspace*; the restriction is what upgrades an implication to an equivalence, and it is the paper's own refinement. The projection-freeness theorem, whose uniqueness half is a genuine and non-obvious argument. And the perfect-interference construction with the one-parameter dial, which is both the sharpness witness for the coherence hypothesis and the clearest exposition of the balance law's content.

Level. The manuscript presents a comprehensively structured theoretical framework. The architecture of the work comprises the bidirectional refutation of a central conjecture, the introduction of an exact replacement identity, a complete characterization of the zero set of the new term, a formal connection to an established object in statistical mechanics, and sharpness witnesses for each hypothesis of the monotonicity theorem. The individual proofs are concise and rely strictly on the foundational Hilbert-space geometry of conditional expectation. Consequently, the originality of the work lies in the structural assembly of these components and the specific selection of the metric that facilitates their integration.

5 Priority in the literature

Finding 6 (Two positioning items are missing)**.** *The Pythagorean half has a close antecedent.* Wolfer and Watanabe, "Geometric aspects of data-processing of Markov chains", *Trans. Math. Appl.* **8**(1) (2024), tnae001, examine data-processing of Markov chains through information geometry, construct a foliation of the lumpable stochastic matrices into exponential and mixture families, and establish a Pythagorean decomposition, $D(P\|P') = D(P\|P_0) + D(P_0\|P')$, relative to a projection onto the lumpable family. The geometric content (an orthogonal splitting relative to a projection onto coarse structure, under a data-processing map on Markov chains, with contraction on one side) is the same content as Lemma 2.3(c) and Theorem 4.2(ii), in a different geometry: a dually flat Kullback–Leibler structure there, a law-weighted quadratic form here, and a decomposition of the chain there against a decomposition of the deficit cochain here. The differences are real and the manuscript's object is not theirs. But the reader who knows that work will want the comparison made, and making it would sharpen rather than dilute the claim: what is new here is the *source term*, which has no counterpart in a Pythagorean identity, and the balance law that pairs it with the transmitted part.

The open-quantum-systems comparison should cite the inequivalence. Section §5.1 identifies production with "information backflow" at the system–environment cut. Chruściński, Kossakowski and Rivas establish that the two standard notions do not coincide: every divisible map is Markovian in the distinguishability sense, while there exist indivisible maps with no information backflow at

all, so backflow is strictly the stronger notion and failure of divisibility does not deliver it. The comparison is worth keeping and worth attributing; it should be labelled an analogy.

The proposed test has an established empirical counterpart. Open problem (iv) proposes a test of $\mathcal{P}_\lambda = 0$, hence of two-time weak lumpability, from data, and observes that "its null distribution is unexplored". Testing whether an observed coarse-grained family composes is routine practice in the validation of Markov state models, where it is performed as a Chapman–Kolmogorov test on a reduced description obtained by exactly the kind of cut this paper studies. Whatever is known there about the behaviour of such a test is the natural starting point, and the open problem would be better posed against it. **[R]**

6 Specific criticisms

Items are marked **[M]** where they bear on the correctness or the substance of a claim, and **[R]** where they concern presentation, attribution or scope.

[M] The metric is claimed to lie in a class that, as written, excludes it
(Finding 1), and Remark 2.4 asserts the opposite two paragraphs later. The conservation identity fails by up to forty-five per cent under the weights of that class. Cite norm equivalence instead, or widen the class.

[M] "Memory" means the two-time composition defect, and the paper's own Proposition 6.5 separates the two
(Finding 4): there the coarse family is exactly flat while the lumped process is not Markov, by exactly $\frac{1}{24}$. Qualify the word throughout.

[M] The abstract claims projection-freeness without the flatness hypothesis
(Finding 2) that Theorem 5.2 states and that Remark 6.7 and Proposition 6.5 show is indispensable; and the uniqueness proved covers only block-supported stochastic sections, not the Mori, Wiener or nonlinear projections whose dependence the formalism actually debates.

[M] The residual at the nearest flat family is not canonical
(Finding 3); the metric projection onto a nonconvex model set need not be unique, and a two-state witness with two minimisers at identical distance is recorded in the verification section. Only h_n is canonical.

[R] The charge is never computed
(Finding 5), although it is the paper's title object; the abstract's "the charge inherits the balance law" overstates what §2.1 concedes, and the quantitative bound offered in Example 3.2 contains a non-constructive constant.

[R] The sentence calling production "non-negative" should be deleted;
it is vacuous, and its connotation is refuted by Proposition 6.5.

[R] The dilation category is never declared,
although equation (3) depends on it.

[R] Two positioning items are missing
(Finding 6): the information-geometric Pythagorean decomposition for data-processing of Markov chains, and the Chapman–Kolmogorov test used in practice to validate reduced Markov descriptions.

[R] "Weak lumpability" is applied to a condition that does not meet its standard definition,
which requires the lumped process to be Markov; the corollary's own proof shows the zero set of production is strictly weaker. *Two-time CK lumpability* would say what is meant.

[R] The identification of production with information backflow runs in the unsupported direction;
indivisibility does not imply backflow, which is the strictly stronger notion. Present it as an analogy.

[R] The plug-in stability claim of open problem (iv) needs a positivity margin
on the coarse block masses and joint propagation of the error in the estimated laws, since the disintegration divides by those masses.

7 Questions to the author

1. The weights (6) depend on both the source and the destination state; the admissible class you cite is written with weights depending on the source alone. Are the two compatible, and if the citation is only to norm equivalence, should it not say so? I find the conservation identity fails by up to forty-five per cent under source-only weights.

2. Theorem 5.2 assumes the fine family is flat, and Proposition 6.5 exhibits a charged fine family for which production is nonzero while the coarse deficit vanishes. Should the projection-freeness claim of the abstract and of §5 not carry that hypothesis?

3. What is $h_2(V^\lambda)$ for Example 3.2? I obtain 0.0265 against a deficit norm of 0.0276. Would a charge column in Table 1 not make the paper's headline quantity concrete, and give the programme its first calibration of the constants in (7)?

4. Can L_*, C_* and θ be bounded for $n = 2$ on a three-point grid? Every quantitative charge statement in the paper and its companion rests on constants that are never estimated.

5. In what sense is production "non-negative"? Proposition 6.5 makes it cancel the transmitted deficit exactly.

6. Which dilation category is declared? Equation (3) holds in the time-inhomogeneous category and fails in the time-homogeneous one.

7. In Proposition 6.5 the coarse family is exactly flat. Is the lumped process Markov? I compute that it is not, by exactly $\frac{1}{24}$. What then is "destroyed entirely" by the interference, and should "memory" not be qualified as two-time composition memory throughout?

8. Can the nearest flat family fail to be unique? Take $n = 2$ with uniform laws and the family (I, I, swap): the flat families with one-step kernels $(J/2, I)$ and $(I, J/2)$ are both at squared distance exactly 2, and their residuals differ. In what sense is $V^\lambda - W^*$ canonical?

9. Theorem 5.2(iii) quantifies over block-supported stochastic sections. Does that class contain Mori's linear projection onto a chosen span of observables, or a Wiener projection? If not, in what sense is the dependence on the projection dissolved rather than side-stepped?

10. What is the standard definition of weak lumpability, and does the zero set of production meet it? Your own Corollary 4.6 shows it does not.

11. Does failure of divisibility imply backflow of information in an operational distinguishability sense? The known relation runs the other way: divisibility implies no backflow, and the converse fails.

12. How does the orthogonal splitting of Lemma 2.3(c) relate to the Pythagorean decomposition established for data-processing of Markov chains in information geometry, where a projection onto the lumpable family plays the role your compression plays here? Is your production term the object that has no counterpart there?

13. Open problem (iv) proposes a test of zero production. How does it relate to the Chapman–Kolmogorov test already used to validate reduced Markov descriptions, and does that practice supply the null behaviour you describe as unexplored?

14. Open problem (i) asks for the geometry deciding the sign of $\langle \Phi_\lambda, \mathcal{P}_\lambda \rangle$. Does the dial of Remark 6.6 generalise: is the production always independent of the component of the fine deficit that is block-measurable, so that the cross term is linear in that component and the sign question is one-dimensional?

15. Can the balance law be stated for the full profile h_d rather than the dimension-matched level, as open problem (ii) proposes, given that the model sets at $d > n$ are closures of compressions rather than zero sets of a polynomial system?

16. Does the severed component Ξ_λ have an interpretation as lost information, and is its norm related to any of the established quantities measuring what a lumping destroys?

8 Technical conclusion

Every presented quantity in this manuscript is correct and has been independently verified. Utilizing exact rational arithmetic, the laws, kernels, deficit matrix, and norm of Example 3.2 were recomputed, along with the exact flatness of Example 3.4, the factorization of Example 3.3, and every matrix, joint law, three margins, four norms, and the cross term of Proposition 6.5. Additionally, evaluations across every composition of a five-point grid confirmed the conditional-expectation lemma, the balance law, the orthogonal conservation identity, the isometry of the block lift, the contraction of the flux, and the discrete Dyson expansion.Manual derivations also confirm the zero set of production, the strong-lumpability computation, the uniqueness argument of the projection-freeness theorem, and the tensor algebra of the autonomy proposition. The manuscript's assertion that it excludes numerical optimization, simulations, and empirical datasets, providing instead manually verifiable quantities, is accurate. This transparency facilitated a comprehensive, rather than partial, audit, representing an exceptionally high standard of reproducibility for a theoretical work.

The manuscript presents a thorough theoretical framework. The architecture of the work includes a bidirectional refutation of the core intuition prior to building the main framework, the establishment of an exact replacement identity, and a comprehensive characterization of the zero set of the new term. Furthermore, it provides a formal connection to an established object in statistical mechanics and a sharpness witness for each hypothesis of the monotonicity theorem. Additionally, the analysis of the distinction between the Chapman–Kolmogorov equations and the Markov property in Corollary 4.6 is both accurate and explicit, successfully resolving a theoretical point where arguments in this literature frequently falter.

Four defects bear on the substance, and none touches a theorem; all four are discrepancies between what is proved and what is announced. The metric foundational to the paper's geometric assertions is nominally assigned to a class that, by definition, excludes it. This structural inconsistency is acknowledged within the manuscript two paragraphs later, where it is noted that the construction fails under generic weights. Quantitative analysis confirms that this operational failure reaches forty-five percent. The word "memory", which carries the significance statement and both destruction results, denotes only the two-time composition defect, and the paper's own sharpest construction separates the two: in Proposition 6.5 the coarse family is exactly flat while the lumped process remains non-Markov by exactly one twenty-fourth, so the interference cut annihilates the accountant and not what it counts. The projection-freeness claim overstates twice, dropping the flatness hypothesis that the theorem states and that Proposition 6.5 shows indispensable, and quantifying only over block-supported stochastic sections while naming the Mori and Wiener projections whose dependence the formalism actually debates. And the residual offered as the canonical geometric representative of the memory kernel is not canonical, since the metric projection onto a nonconvex model set need not be unique: I exhibited a two-state family with two flat families at identical distance and different residuals.

The remaining items are matters of scope. The charge, which is the paper's title object and the subject of its abstract, is never computed; every quantitative statement is about the deficit cochain, and the transfer to the charge runs through constants that are non-constructive throughout this programme. I supply the missing number for Example 3.2. The dilation category is not declared although one identification depends on it. A sentence calling production "non-negative" should go. The classical term "weak lumpability" is applied to a condition the paper itself proves to be strictly weaker than it, which is the one place where its otherwise exemplary handling of the composition-versus-memorylessness gap lapses; the identification of production with information backflow runs against the known relation between those two notions; and three positioning items are absent: an information-geometric Pythagorean decomposition for data-processing of Markov chains, which is the nearest antecedent of the conservation identity; the inequivalence of divisibility and backflow; and the Chapman–Kolmogorov test already used in practice to validate reduced Markov descriptions, which is the empirical counterpart of the test proposed among the open problems.